

Technology Stack

SMARTBRIDGE / SKILL WALL

Date: 29 September 2026

Team ID: SWTID-2026-3580

Project Name: EduGenie: Google Gemini Powered Learning Assistant

Maximum Marks: 2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that will be used to build the solution. The selected stack supports maintainability and alignment with EduGenie's learning-assistant requirements.

The table below lists the technologies selected for each architecture component and their purpose.

Technology Stack Details			
S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML + CSS + JavaScript	Provides the responsive web interface for selecting learning tasks, entering requests, and

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2	Backend / Server-Side	Python + FastAPI	Provides REST endpoints, request validation, routing, and integration with the AI modules.
3	Database / Data Storage	No database in MVP	The current MVP does not require persistent database storage; learning requests are processed directly by the application.
4	Cloud / Hosting / Deployment	Uvicorn + Local Deployment	Uvicorn runs the FastAPI application locally for development, testing, and demonstration.
5	Version Control & CI/CD	GitHub + Git	Stores the source code and project deliverables, supports version history, and enables team collaboration.

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S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
6	Third-Party APIs / Other Tools	Google Gemini API + LaMini-Flan-T5-783M	Gemini provides Q&A, quiz, summarization, and learning-path generation; the local LaMini model supports concept explanation.

Selected Stack Summary

Frontend: HTML, CSS, JavaScript | Backend: Python, FastAPI | AI: Google Gemini API + LaMini-Flan-T5-783M
Deployment: Uvicorn/local development | Version Control: Git/GitHub | Database: Not required for the MVP